Fire Detection & Evacuation

D7065E, Architecture Document (final report)

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1 Summary

The system detects fires in every room and corridor of a three-floor university building, tells real fires apart from cooking, steam, dust, smoking, heaters and broken sensors, releases the sprinkler only where it burns, unlocks the access-controlled doors for egress while closing the fire doors, purges the smoke, predicts which zones the fire will reach next and evacuates the occupants floor by floor along routes that avoid fire, smoke, predicted spread and impassable doors. The building, its rooms, doors, equipment and 3D view are BuildSim's; this project provides the physics, the devices, the intelligence and the control around it. The thread through every decision is *a learned model may make the system faster, never less safe*: fixed, verifiable rules and a 90 °C safety net always stand behind it.

Field	Value
Project title	Fire Detection & Evacuation
Team members	C. Lisper
Use case	Fire detection, suppression, egress and evacuation (956 zones, 3 floors)
Repository	https://github.com/Thekingofswords/d7065e-fire-response; services in `devices/`, deployment in `compose.override.yaml`
Version / date	v2.0 / 2026-10-10 (all plan items done, live run on the real BuildSim)

Changes since the proposal. Five design changes were made because the evidence demanded them. (1) *Messaging*: the proposal used REST between our processes; 956 zones × 4 signals every 2 s with four consumers (agent, database, display bridge, tests) made pub/sub the obvious fit, so the design moved to MQTT, which also gave crash detection (last wills) and restart recovery (retained values) for free. (2) *Models*: the proposed separate model service became in-process detectors, because one call per tick for all zones takes milliseconds and a network hop on the safety path is a failure point. (3) *Physics*: the proposed linear diffusion made false-alarm rejection trivial (smoke never reached a neighbour realistically), so it was replaced by a zone fire model with doors, corridor cells and stairwells. (4) *Doors*: the proposal unlocked doors on routes; doors are now locked by an access-control schedule in normal operation, every one is released at a fire alarm, and fire doors close (not lock) to contain smoke. (5) *Evacuation*: the agent now owns the evacuation plan (when, which floors, all-clear), while a separate service routes individual people. Each change is revisited where it matters below.

2 Use case and building context (BuildSim)

The building. The system protects the BuildSim model of a university building: three floors, 956 rooms and corridors with sensors, 819 doors and the walkable graph between them. BuildSim is the *shared state* of the cyber-physical system. It stores the latest room layers, equipment, doors, alerts and viewer sessions, exposes a REST API to read and write them, and streams the 3D view to the browser over WebSocket. It does not simulate physics, decide anything or keep history. **BuildSim is not modified; it is instrumented**, which keeps the seam between the physical half (behind BuildSim's API) and the cyber half (our processes) clean enough that the same software could drive a real building.

How the system uses BuildSim.

- The **fire simulator** is the only process that touches the environment: it reads the actuator and door states back from BuildSim every tick (`GET /api/equipment?category=fire_suppression`, `GET /api/doors`), advances the physics and writes the truth as room layers (`PUT /api/room-layers`) and visual effects.

- The **sensor gateways** (one per floor) read those layers, add the behaviour of real detectors (lag, noise, faults) and publish readings to the broker. They emulate devices; they never create the environment.
- The **actuator gateway** and the **door gateway** apply the agent's commands to BuildSim (`PUT /api/actuators/{id}/state`, `PUT /api/doors`) and confirm only what BuildSim accepted. The control loop therefore closes *through* BuildSim: fire → layers → sensors → agent → actuator → BuildSim → fire.
- The **occupants** service and the **display bridge** use BuildSim's entities, session highlights, routes and coverage zones to show the people, the hazard colours and the smoke in the viewer.

Actors. The *building manager* watches the BuildSim viewer and Grafana and may command an actuator or a door by hand (the agent adopts a sprinkler it did not start). *Occupants* follow a daily schedule; they are sensed (presence, CO_2) and evacuated, but are not on the control path.

The scenario the demo shows. At 10:00 on a Monday, a waste-paper basket smoulders in office A109 for two minutes and then flames as a fast t^2 fire; half of the doors stand open and 38 people are in the building. Without the system A109 flashes over after about five minutes and smoke fills the corridor and, through the stairwell, the floor above. With it, the smoke alarm in A109 releases the access control, closes the floor's fire doors and starts the evacuation of all three floors; about 30 s after flaming begins the heat detector's rate of rise, confirmed by CO, releases the A109 sprinkler and the fire dies within a minute. The corridor never exceeds 0.5 %/m of smoke (44 %/m with the doors left as they were), the purge damper clears A109, and the agent gives the all-clear. At night the same alarm unlocks the more than 500 office and stairwell doors the schedule had locked.

Why it matters. Most fire deaths are caused by smoke, not flames, and a fire grows from a basket to flashover in minutes. Detection must be fast, must not flood a building with water for every burnt toast (false releases are expensive and erode trust), must survive a broken sensor in exactly the room that burns, and must get people out through doors that are locked for security the rest of the time. Those four concerns, *speed, discrimination, resilience and egress*, drive the requirements in Section 3.

3 Requirements

Each requirement is testable, has a unique ID, names the design element that satisfies it (Sections 4–9) and the test or experiment that verifies it (Section 10). FR = functional, NFR = non-functional, SEC = security, SAF = safety. Times are simulated seconds unless stated; the simulation runs at 5× real time.

ID	Requirement	Prio.	Acceptance criterion	Verified by
FR-01	Release the sprinkler in a zone with a flaming fire	Must	$\geq$ 95 % of fires released $\leq$ 60 s after flaming	`baseline.py`; `test_fire_is_detected_and_suppressed_through_buildsim`
FR-02	No release where there is no fire (nuisances, smoke from next door, faulty sensors)	Must	$\leq$ 1 false release per 100 zone-hours	`baseline.py`; `train_model.py`; `test_hot_smoke_from_next_door_is_attributed_not_released`
FR-03	Stop the water once the fire is out	Must	After $\geq$ 300 s of water, once $\leq$ 35 °C and not heating	`test_sprinkler_runs_its_minimum_time_then_stops_when_cool`

ID	Requirement	Prio.	Acceptance criterion	Verified by
FR-04	Purge smoke from every zone it reaches	Should	Damper opens ≥ 3 %/m, closes ≤ 0.5 %/m; < 1 %/m within 15 min of suppression	`test_smoke_purge_*`, `response.py`
FR-05	Alert the building manager	Must	One alert per zone in alarm or suppression, most severe first, ≤ 100; one per offline gateway	e2e fire and gateway-crash tests
FR-06	Route occupants out avoiding fire, smoke and impassable doors	Should	Detour where an alternative exists; re-plan on a jammed or locked door	`test_jammed_door_makes_the_evacuation_replan`
FR-07	Close fire doors on an alarm, recall lifts on a fire	Should	Within one tick; closed but unlocked; normal 120 s after the last alarm	`test_fire_alarm_closes_doors_and_recalls_lifts`
FR-08	Enforce door safety rules whoever commands	Must	Exits never locked or blocked; jammed doors not operable; unknown sources refused	`test_door_rules_are_enforced`
FR-09	Realistic occupancy over the day	Should	Arrivals 07–09, lunch dip, departures 16–18, empty nights and weekends	`test_schedule.py`
FR-10	Unlock access-controlled doors for egress on any alarm	Must	Out of hours > 85 % of locked doors released within one tick; exits never locked	`test_fire_alarm_unlocks_egress_doors`, live Run B
FR-11	Detect anomalies against normal behaviour; classify the cause	Should	≥ 98 % of fires and nuisances anomalous ≤ 60 s; ≤ 1 false anomaly / 100 zone-h; non-fire anomaly $\rightarrow$ “check zone”, no water	`train_anomaly.py`, `test_anomaly.py`, live Run C
FR-12	Predict fire spread	Should	AUC ≥ 0.85 and better than the neighbour baseline on held-out data	`train_spread.py`, `test_spread.py`
FR-13	The agent orchestrates the evacuation	Should	One versioned plan per change; phased floors never shrink; all-clear only from the agent	`test_plan_is_phased_*`, `test_agent_plan_drives_the_evacuation`
FR-14	Dashboard as specified	Should	Red fire, orange risk/smoke, green safe; smoke coverage; route; every update accepted by BuildSim	`test_dashboard_highlights_and_coverage`, live Run A
FR-15	Evacuation routes consistent with BuildSim’s route API	Should	Without hazards, local and BuildSim routes agree within 5 %	`test_routes_agree_with_buildsims_route_api`
NFR-01	Decide for the whole building every tick	Must	Agent step ≤ 0.5 s for 956 zones (interval 2 s)	`scaling.py`
NFR-02	Survive a sensor-gateway crash	Must	Offline alert ≤ 30 s; readings back ≤ 30 s after restart	e2e crash test, live Run A
NFR-03	Survive an agent restart without stopping water	Must	A running sprinkler stays on	`test_agent_restart_keeps_a_running_sprinkler_on`
NFR-04	Survive a BuildSim restart	Must	Equipment, states and doors restored ≤ 30 s	`test_buildsim_restart_is_recovered`, live Run A
NFR-05	Never act on corrupt data	Must	Malformed, non-numeric, impossible, future, duplicate readings rejected and reported	`test_messages.py`, e2e corrupt-data test

ID	Requirement	Prio.	Acceptance criterion	Verified by
NFR-06	Apply commands quickly	Should	Command → applied and confirmed ≤ 1 s (p90)	`latency.py` (live)
NFR-07	Detect a fire with a broken sensor in the fire room	Should	Frozen or offline thermometer: fire still released, no flashover	`baseline.py --faults`
NFR-08	Degrade safely on bad inputs	Must	Degraded ≤ 10 s, critical alert, never switch water off, recover automatically	`test_degraded_mode`, live Run A
NFR-09	Stop cleanly, restart when hung	Should	SIGTERM → "offline" published; hung 120 s → exit and restart	`test_graceful_shutdown`, `test_lifecycle.py`
NFR-10	Keep history cheaply	Should	Raw compressed after 1 day, kept 30 days; per-minute aggregate; days archived to Parquet	live Run A (TimescaleDB jobs)
NFR-11	One clock	Should	All services follow `sim/clock`; no jumps or backwards time	`test_clock.py`
SEC-01	Only authorised services may command	Must	No login refused; a sensor login cannot command a sprinkler	`test_broker_enforces_users_and_acl`
SEC-02	Encrypted broker, secrets outside the repository	Should	Verified TLS; plain MQTT and unknown CA refused; passwords from `.env`	`test_tls.py`, live Run D
SAF-01	A learned model is never the only thing between a fire and water	Must	A healthy thermometer at ≥ 90 °C releases regardless of detector	`test_safety_net_releases_even_when_the_detector_says_no`

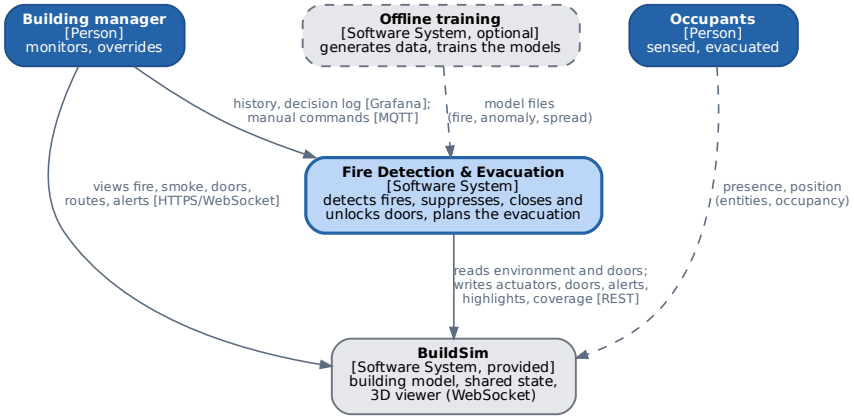
Two tensions run through these requirements. Speed (FR-01) pulls towards releasing early; discrimination (FR-02) pulls towards waiting for evidence. Resilience (NFR-07) pulls the same way as speed: a room whose thermometer is broken must still get water, which costs precautionary releases. Resolving these safely is the autonomous service's whole job (Section 7), and it is why neither a single threshold nor a learned model alone is enough.

Traceability. The five viewpoints map onto the document: context (4.1), functional (4.2–4.4), information (5–7), behavioural (8) and deployment (9). Section 10 closes the requirement → design → validation thread.

4 Architecture

4.1 Context diagram (C4 Level 1)

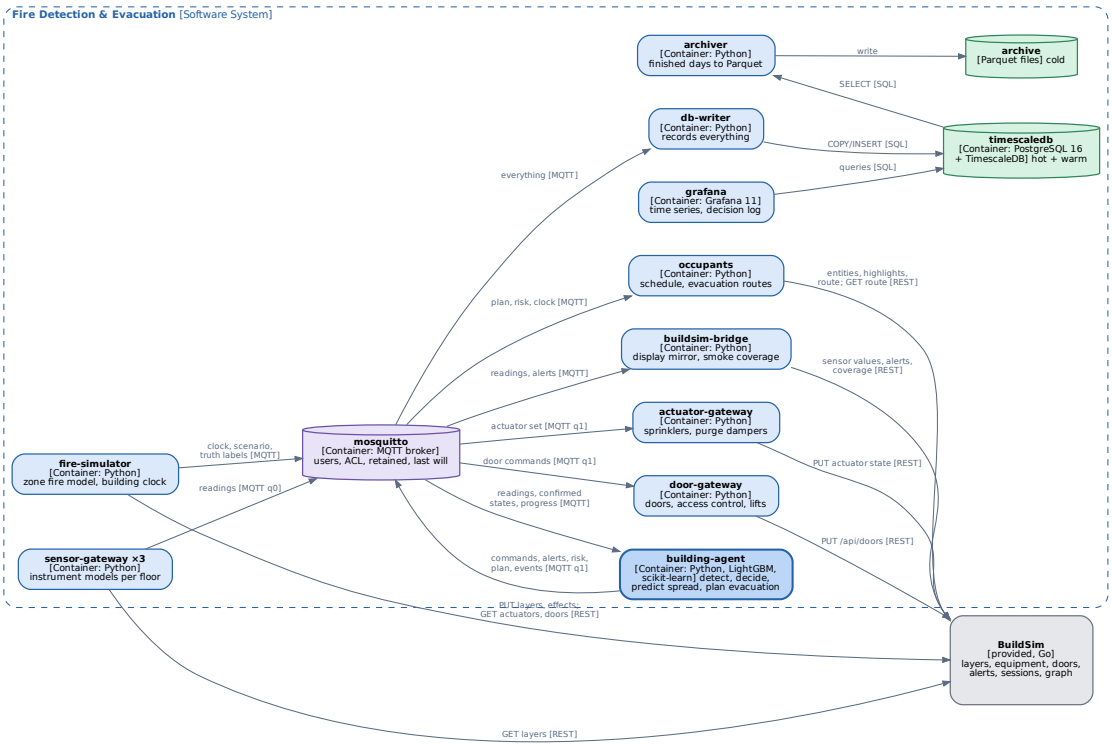
The system reads and commands the building only through BuildSim. The building manager monitors the viewer and Grafana and may override through MQTT; occupants are sensed and evacuated, never connected; model training is an offline system that hands over model files and is never on the control path.



C4 System Context (Level 1). The system in scope is highlighted, external systems are grey; every arrow is labelled with its purpose and protocol.

4.2 Container diagram (C4 Level 2)

Every box is one service in `compose.override.yaml` and one file in `devices/` (the occupants service is `examples/api/scenario/populate_evacuation.py`). The arrows carry the protocol. The fire simulator interacts only with BuildSim and the broker; the sensor gateways read what it wrote; the agent decides; the gateways apply its commands to BuildSim; the simulator reads them back next tick, which is the closed loop that makes this a cyber-physical system rather than a dashboard.



C4 Container diagram (Level 2). Each container shows its technology and responsibility; the dashed boundary is the system in scope, BuildSim is grey.

4.3 Component responsibilities

Every container has one job, which is what makes it testable and lets one part fail without taking the rest down. The last column is the systems reasoning: why the job is not merged into a neighbour.

Container	Responsibility (one job)	Tech	Reason for separation
fire-simulator	Advance the physics of every active zone; write truth; read actuators and doors back; publish the building clock	Python	It *is* the environment; nothing on the decision side may share its memory
sensor-gateway ×3	Turn the environment into realistic, faulty readings for one floor	Python	A crash darkens one floor, not the building; floors scale out independently
mosquitto	Decouple producers from consumers; retain last values; last wills; enforce who may publish what	Mosquitto 2.0	One source, many consumers; one place for command authority
building-agent	Detect, decide sprinklers, dampers, doors and lifts; predict spread; own the evacuation plan	Python, LightGBM, scikit-learn	The safety-critical logic, isolated from everything that can fail around it
actuator-gateway	Validate sprinkler and damper commands, apply to BuildSim, then confirm	Python	"Deciding is not acting": a wrong command is rejected independently of the agent

Container	Responsibility (one job)	Tech	Reason for separation
door-gateway	Own BuildSim's door collection: access schedule, door rules, full snapshot, confirmations	Python	`PUT /api/doors` replaces the collection, so exactly one writer; exit rules hold whoever commands
occupants	People with a daily schedule; follow the plan; routes, highlights, progress	Python	Not on the control path; can fail or be left out without affecting fire control
buildsim-bridge	Register detectors; mirror readings, alerts and smoke coverage into BuildSim	Python	BuildSim takes one request per value; rate limiting and deadbands in one place, off the control path
db-writer	Record readings, labels, commands, states, events, scenarios; set up storage tiers	Python, psycopg	Storage may be slow or down without affecting control
timescaledb	Hot and warm history, training data, audit trail	PostgreSQL 16 + TimescaleDB	The only durable record; BuildSim keeps nothing
archiver	Copy finished days to Parquet (cold tier)	Python, pyarrow	Batch work every 6 h, independent of recording
grafana	Time series, status tiles, decision log	Grafana 11	Presentation, read-only on the database

4.4 Design decisions and justifications

Each row follows "X because (requirement); the alternative was Y; the trade-off accepted is Z".

Decision	Chosen	Alternatives	Justification (requirement)	Trade-off
Sensor transport	MQTT pub/sub, QoS 0 retained	REST to each consumer; Kafka	Four consumers need every reading; retained values and last wills give restart and crash detection (NFR-02, NFR-03)	A broker to run; at-most-once readings (replaced 2 s later)
Loop through BuildSim	Simulator writes layers and reads actuators from BuildSim	Simulator ↔ sensors directly	BuildSim is the shared state; the same software could drive a real building behind the same API	Two REST reads per tick; BuildSim on the critical path (NFR-04)
Device granularity	One gateway per floor, one actuator gateway	One process per device	956 zones × 6 devices would not fit a laptop; real buildings use gateways (NFR-01)	A crash darkens a floor (one alert per gateway)
Detection	Rules by default; LightGBM or hybrid optional; all behind a 90 °C safety net	Rules only; deep model; LLM	Rules are verifiable and survive broken sensors (FR-01, NFR-07); models cut false releases and release earlier (FR-02); the safety net bounds model failures (SAF-01)	Models must be retrained when the physics changes
Anomaly stage	Isolation forest + autoencoder on normal data; cause by a supervised classifier	One-class SVM; autoencoder only	Two notions of "unusual" (isolation, reconstruction) catch more; scikit-learn only (FR-11)	A second model file, version-pinned

Decision	Chosen	Alternatives	Justification (requirement)	Trade-off
Spread prediction	LightGBM on adjacency-aggregated features weighted by door openness	Graph neural network; physics look-ahead	Adjacency plus temperature gradients as asked; milliseconds per tick; explainable (FR-12)	Learns the simulator's physics
Features in the agent	`features.py` shared by training and the agent	Pipeline service	No train/serve skew; no extra hop (NFR-01)	~6 min of history per zone in memory
Command reliability	QoS 1 + separate confirmed state + re-send	Fire-and-forget; REST response	The agent acts on what was applied, not what it asked for (FR-01, NFR-06)	Two topics per actuator
Physics fidelity	Zone fire model with cells, doors, stairs	Linear diffusion; CFD	Smoke must drift into non-burning rooms realistically, or discrimination is trivial (FR-02)	More parameters; still a simplification
Doors	One door gateway owns the snapshot; commands per door, zone, floor, lifts, egress	Each service writes doors	`PUT /api/doors` replaces everything; rules in one place (FR-08, FR-10)	Changes batched ($\leq$ 1 s)
Fire-door policy	Close (not lock) the alarmed floor's fire doors	Alarmed zone only; lock	Contains smoke before the fire is confirmed (FR-07), keeps every door an escape route (FR-06)	Doors close for nuisance alarms too
Evacuation ownership	Agent owns phase, floors, all-clear; occupants route people	Agent routes everyone; occupants decide alone	Deciding to evacuate is safety-critical, routing individuals is not (FR-13)	Two processes must agree (versioned, retained plan)
Evacuation routing	Local hazard-weighted Dijkstra, cross-checked with `/api/graph/route`	BuildSim's route API only	BuildSim's route cannot weigh fire, smoke or risk (FR-06, FR-15)	Two route computations to keep consistent
Storage	TimescaleDB hot + warm, Parquet cold	Plain PostgreSQL; files only	Fast recent queries and cheap history (NFR-10)	Two places to look
Command authority	Mosquitto users + ACL + gateway validation; optional TLS	Shared secret in payloads	Enforced before any consumer sees a message (SEC-01, SEC-02)	Passwords and certificates to manage
Lifecycle	Shared `lifecycle.py`: status topic, heartbeat, watchdog	Docker restarts only	A degraded service says so and acts safely (NFR-08, NFR-09)	A little code in every service

5 Interfaces and communication

Three patterns are used, each for the job it fits. **Publish–subscribe (MQTT)** carries readings, commands, confirmations, alerts, the plan and events between our processes, because one source feeds several consumers and the broker buffers a briefly absent one; QoS 1 and a separate retained confirmation topic make commands reliable. **Request–response (REST)** is used for every call to BuildSim, because each is a question or a state change whose result the caller must check. **WebSocket** (BuildSim's own) pushes the 3D view to the browser. SQL connects the database to Grafana and the archiver. Room names are URL-encoded in topics (`A103+A112_B+A103` → `A103%2BA112_B%2BA103`), since `+` is an MQTT wildcard.

A reading and a command, as they cross the bus:

```
building/level0/A109/sensors     (QoS 0, retained, every 2 s per zone)
{"ts": 1791468036.2, "gateway": "sensors-level0",
 "temperature": 31.4, "smoke": 9.8, "co": 42.0, "co2": 655.0}

building/level0/A109/actuator/sprinkler/set      (QoS 1, re-sent until confirmed)
{"state": "on", "reason": "SUPPRESSING: rate of rise",
 "id": "9cac88124bbe", "ts": 1791468041.8, "source": "building-agent"}
```

Units: temperature °C (−40…1200), smoke obscuration %/m (0…100), CO and CO_2 ppm (0…10 000 / 100 000), `ts` seconds since the epoch. Every consumer validates every reading (`messages.py`); anything outside these ranges, not numeric, more than 60 s in the future or not newer than the last accepted one is dropped and counted (`data_quality` events).

Topic	Producer → consumers	Payload	QoS / retain / rate
`building/<lvl>/<room>/sensors`	sensor gateway → agent, db-writer, bridge	ts, gateway, temperature, smoke, co, co2	0 / yes / 2 s per zone
`…/actuator/<name>/set`	agent, operator → actuator gateway	state (`on`/`off`, damper 0–1), reason, id, source	1 / no / on change, re-sent 10 s
`…/actuator/<name>/state`	actuator gateway → agent, db-writer	state, ts, command_id, reason, source, corrected	1 / yes / after BuildSim accepted
`building/<lvl>/doors/set`, `…/<room>/doors/set`, `building/lifts/set`, `building/doors/egress/set`	agent, occupants, operator → door gateway	state and/or lock_state, reason, id, source	1 / no / on change
`building/doors/<id>/state`	door gateway → agent, occupants, db-writer	state, lock_state, kind, level, room, reason, source	1 / yes / on change
`building/alerts`	agent → bridge	list of {id, severity, title, message, level, room}	1 / yes / on change
`building/risk`	agent → occupants, db-writer	zones {zone: p}, horizon_s 300, model, threshold	1 / yes / on a new set or 0.1 change
`building/evacuation/plan`	agent → occupants, db-writer	phase, evacuate_levels, fire/alarm/smoke/risk zones, exits, doors, version, clock	1 / yes / on change
`building/evacuation/progress`	occupants → agent	present, pre_movement, evacuating, evacuated, awaiting_rescue {zone: n}, route_checks	1 / yes / 5 s

Topic	Producer → consumers	Payload	QoS / retain / rate
`building/gateways/<name>/status`, `building/services/<name>/status`	every service	status, lifecycle state, detail, since (last will: offline)	1 / yes / on change
`sim/clock`, `sim/scenario`	simulator → all	sim_time, week_minute, "Mon 07:31"; scenario and fault schedule	0–1 / yes / every tick
`sim/truth/<lvl>/<room>`	simulator → db-writer only	true values + labels (training and evaluation only)	0 / yes
`building/events`	all → db-writer	ts, type, severity, level, room, message, …	1 / no

Collection `PUT`s in BuildSim replace everything, so each has exactly one writer:

BuildSim endpoint	Sole writer	Readers
`PUT /api/room-layers`, `/api/effects`	fire-simulator	sensor gateways, viewer
`POST /api/equipment/bulk`, `PUT /api/sensors/{id}/value`, `PUT /api/alerts`, `PUT /api/coverage`	buildsim-bridge (≤ 150 requests/s, deadbanded)	viewer
`POST /api/equipment/bulk` (`fire_suppression-…`), `PUT /api/actuators/{id}/state`	actuator-gateway	fire-simulator (`GET /api/equipment`), viewer
`PUT /api/doors` (full snapshot); `door_lock` equipment	door-gateway	fire-simulator, occupants, BuildSim's own routing, viewer
`PUT /api/entities`, `/api/occupancy`, `/api/sessions/{id}/highlights`, `…/route`	occupants	sensor gateways (people for CO_2), viewer

Every decision message carries a human-readable *reason* (`"rate of rise; smoke 30.0 %/m, CO 96 ppm"`), so every action is auditable afterwards, in the alert list and in Grafana's decision log.

6 Simulating the sensor values (the physical model)

BuildSim provides the rooms but not their physics, so a **zone fire model** was built from standard fire-engineering correlations. The aim is not CFD accuracy but enough realism that the detection problem exists: heat, smoke and CO must rise in the right order, drift into rooms that are not burning, get thinner with every doorway and respond to sprinklers, dampers and doors. Each *cell* (a room, or a ~4 m segment of a corridor along its walkable graph) has a hot upper layer and a cool lower layer; only active cells are integrated, so a quiet building costs almost nothing.

Fire. After an optional smouldering phase (little heat, smoky and CO-rich), the heat release grows as a t^2 fire, capped by the fuel and by the air the openings let in (Kawagoe):

$$\dot{Q}(t) = \min(\alpha t^2,\ q'' A_{\text{floor}},\ 1500\, A_o \sqrt{H_o})\ \text{kW},$$

with α from the SFPE growth classes, $q'' = 250$ kW/m² and A_o, H_o the open area and height of the doorways. Flashover occurs at a 600 °C layer; after 70 % of the fuel load the fire decays.

Heat. The hot-layer temperature follows the MQH correlation with a thermal lag, and the detector sees the Alpert ceiling jet on top:

$$\Delta T_{\text{MQH}} = 6.85 \left(\frac{\dot{Q}^2}{A_o \sqrt{H_o}\, h_k A_T} \right)^{1/3}, \qquad \Delta T_{\text{jet}} = \begin{cases} 16.9\, \dot{Q}^{2/3} / H^{5/3} & r/H \leq 0.18 \\ 5.38\, (\dot{Q}/r)^{2/3} / H & \text{otherwise.} \end{cases}$$

Smoke and gases. A Heskestad plume, $\dot{m} = 0.071\,\dot{Q}_c^{1/3} z^{5/3}$, fills the ceiling layer, which descends. Soot, CO and CO_2 yields per kilogram of fuel rise when the fire is short of air; soot becomes obscuration through the specific extinction coefficient (8.7 m^2/g flaming, 4.4 m^2/g smouldering). Once the layer is below a door soffit it spills through the door as a buoyant flow,

$$\dot{V} = \frac{C_d}{3}\, w\, h \sqrt{\frac{g\, h\, \Delta T}{T_{\text{cold}}}}, \qquad C_d = 0.7,$$

entraining room air on the way, so smoke is cooler and thinner after every doorway. Closed doors leak 1 % of their area; stairwells carry hot smoke up with a stack flow; background ventilation (2 ACH), soot settling and the purge damper (10 ACH when open) remove it. Sprinklers make the heat release decay as $e^{-t/43\,\text{s}}$ (NIST), much less effectively after flashover, and do not remove smoke; switching the water off early lets the fire regrow.

Look-alikes. Cooking (smoke, some heat, a little CO), steam (dense aerosol, no CO), dust (short burst), heaters (heat only) and smoking (light smoke with CO) create the false-alarm cases the detectors must reject.

Sensors. The gateways model real devices: a heat element with a response time index, $\mathrm{d}T_d/\mathrm{d}t = \sqrt{u}\,(T_g - T_d)/\text{RTI}$ (RTI 50, u the ceiling-jet velocity); a photoelectric chamber that smoke must drift into (lag L/u); first-order CO and CO_2 cells; noise, resolution and range. Faults belong to the device: a drifting smoke chamber, a frozen thermometer, smoke spikes, a dead CO cell, or a device that stops reporting.

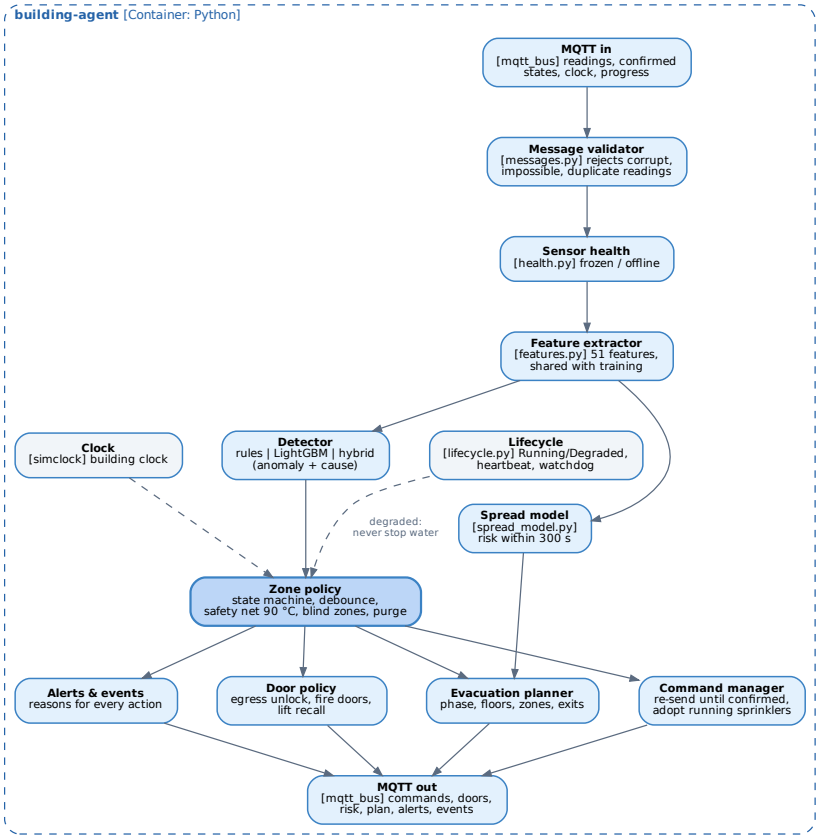
Occupancy. People are agents with a working day on the building clock: an office each, arrival 07:00–09:00 (centred on 08:00) through an entrance, lunch 11:30–12:30 (half in the floor's lunch room, half out), visits to meeting rooms and colleagues along BuildSim's walkable graph, departure 16:00–18:00, nights and weekends empty. Room types come from the floor plan (rooms ≥ 25 m^2 are meeting rooms, the largest per floor is the lunch room). People are BuildSim entities, so the sensor gateways count them for CO_2, and they are the population the evacuation must move.

The loop closes through BuildSim. The simulator reads the sprinkler, damper and door states back from BuildSim every tick: water makes the heat release decay, an open damper drains smoke, a closed door leaks only 1 %. That is why the evaluation in Section 11 can compare policies on the same fire. The model's limitations (well-mixed cells, no wind or HVAC network, fuel per floor area rather than per object) are listed in Section 13.

7 Autonomous services and data engineering

7.1 The autonomous service

One agent decides for the whole building every 2 s. A building-level agent is needed because the decisive evidence is often next door: a rate of rise in a room whose neighbour is hotter and heat-alarmed is smoke spilling in, not a fire, which a per-room device cannot know.



C4 Component diagram (Level 3), inside building-agent. The zone policy sits between every detection and every command.

Detectors (`DETECTOR=rules|model|hybrid`).

- *Rules*: smoke ≥ 8 %/m or CO ≥ 50 ppm raises the alarm; ≥ 68 °C, or a rise ≥ 8.3 °C/min with evidence of combustion (CO ≥ 10 ppm, a rise ≥ 16.6 °C/min, or ≥ 50 °C), releases water, unless a neighbour is hotter and heat-alarmed. Without a working thermometer, dense smoke plus CO releases.
- *Model*: a LightGBM fire probability from 51 features: own signals, rates over 30/60/120/300 s, excess over baseline, maxima, spread, CO per smoke, neighbour maxima and differences, sensor-health flags and a corridor flag. The release threshold (0.85) is the most sensitive one within 1 false release per 100 zone-hours on validation scenarios.
- *Hybrid*: the rules, plus an anomaly stage and a cause stage (7.2). An anomaly classified as fire raises the alarm and releases at P(fire) ≥ 0.98; any other anomaly gives a "Check zone" alert, never water. The rules' decisions always stand.

Why rules are the default. The simplest approach that meets the requirement wins: rules meet FR-01 and FR-02 in the closed loop, survive broken sensors through explicit fallbacks, and every decision can be explained in one sentence. The learned detectors release earlier and halve the smoke (Section 11), so they are offered behind the same safety net, by choice.

Policy. A per-zone state machine (Section 8.3) with two-tick debouncing: the sprinkler runs at least 300 s and stops once the zone is ≤ 35 °C and not heating; the purge damper opens at 3 %/m and closes below 0.5 %/m. On any alarm the agent releases the access control (`building/doors/`

`egress/set`), closes the floor's fire doors (closed, unlocked) and starts the plan; on a confirmed fire it recalls the lifts.

Spread prediction and the evacuation plan. Each tick the spread model estimates, per zone, the probability of becoming untenable (burning, ≥ 60 °C or ≥ 10 %/m) within 300 s; risk is reported within three links of an alarm. The plan goes `normal → evacuate → all_clear`: the alarm floor and the floors above first, every floor on a confirmed fire or after 300 s, never fewer during one evacuation, all-clear 180 s after the last alarm. Occupants report progress back; people sheltering or trapped become a critical alert.

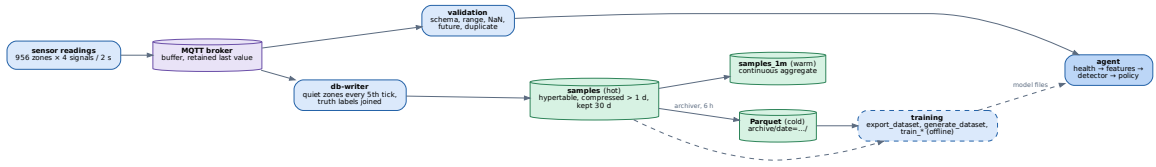
Failure and safety behaviour. A healthy thermometer at ≥ 90 °C releases whatever the detector says (SAF-01). A faulty thermometer can never declare a fire out (frozen readings look cool and steady): the agent uses falling CO and smoke and doubles the minimum discharge. A zone whose fire sensors are all faulty gets water as a precaution when a neighbour shows ≥ 2 %/m smoke or ≥ 15 ppm CO, because closed fire doors otherwise hide its fire until it flashes over (found by `baseline.py --faults`). In *Degraded* mode (a gateway offline or > 20 % of zones stale) water is never switched off. After a restart the agent adopts any sprinkler that is confirmed running; if the model files cannot be loaded, it falls back to the rules.

7.2 The learned models

All three are trained offline on scenarios generated by the same physics, sensors and agent policy as the live system (`generate_dataset.py`, 1,250 random scenarios, ~2.1 million zone-ticks), split *by scenario* 70/15/15 so neighbouring rows of one fire never leak between training and test.

- **Fire model** (`train_model.py`): binary LightGBM; held-out: 90 % of burning zones released, p90 20 s after flaming, 0.97 false releases per 100 zone-hours.
- **Anomaly detection** (`train_anomaly.py`, `anomaly.py`): an isolation forest and an MLP autoencoder ($38 \to 16 \to 8 \to 16 \to 38$) trained on *normal* periods only (before anything happens in a scenario). Each score is divided by its own threshold, set on validation normal data for ≤ 1 false anomaly per 100 zone-hours; the anomaly score is the larger. A four-class LightGBM (normal, fire, nuisance, fault) on the 51 features plus the anomaly score names the cause.
- **Spread model** (`train_spread.py`, `spread_model.py`): LightGBM on 20 features aggregated over the adjacency graph: own temperature, rate, smoke and CO; per neighbour the temperature gradient and smoke, weighted by how open the connection is (open door 1, stairwell 0.5, closed door 0.03, from the door gateway's confirmed states); neighbouring fire and alarm; graph distance to the nearest fire. Labels come from each zone's future in 800 closed-loop scenarios. Baseline: "next to a fire or a markedly hotter zone".

7.3 The data pipeline



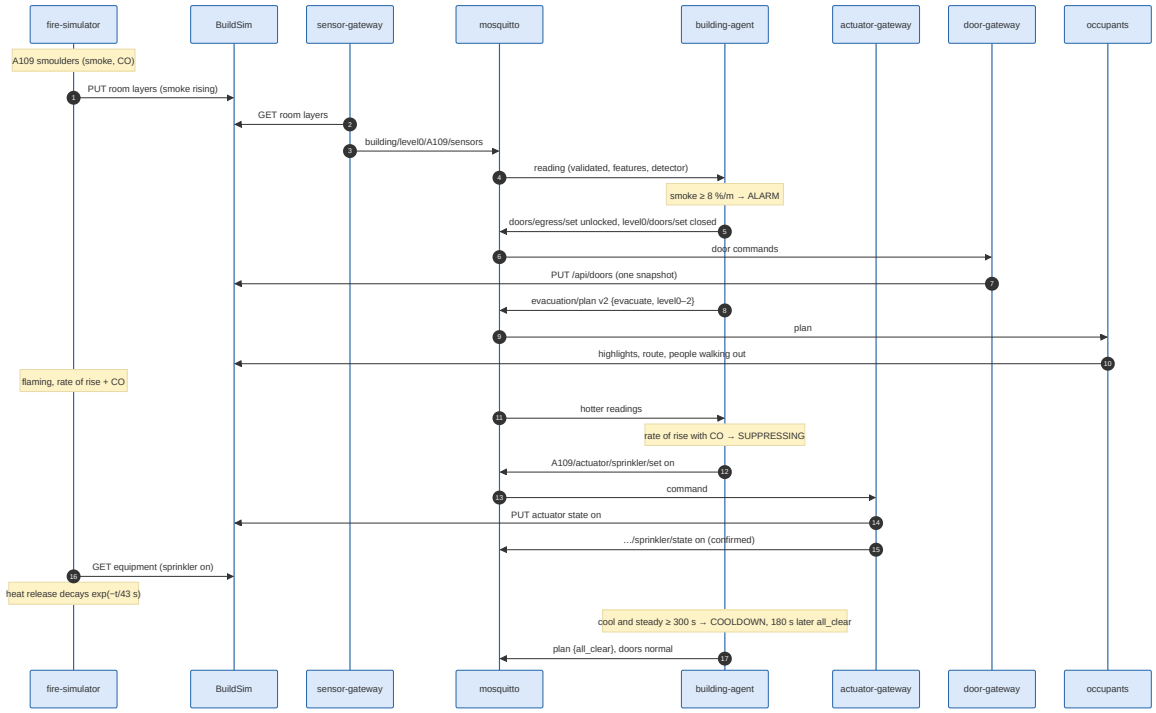
The data pipeline: the broker buffers, the agent consumes live, the database keeps hot and warm tiers, finished days go to Parquet, and training reads both.

Ingestion. The broker is the buffer: if the database is slow, db-writer falls behind and catches up; control is unaffected. Quiet zones are recorded every fifth tick, busy ones every tick (≈ 450 rows per 2 s during a fire). The simulator's truth labels are joined into the same rows, so live recordings and generated data share one schema. **Storage.** TimescaleDB's `samples` hypertable is the hot tier (compressed after a day, kept 30 days), the `samples_1m` continuous aggregate the warm tier (Grafana's floor panels), and Parquet files per day the cold tier for training and analysis (DuckDB, pandas). **Features.** `features.py` computes rates, baselines, spreads and neighbour context, the same code in training and in the agent, so there is no train/serve skew. **Data quality** is treated as a control input, not hygiene: corrupt, impossible, future and duplicate readings are rejected at every consumer and reported; a sensor whose BuildSim timestamp is older than 20 s is *offline*; a reading that has not changed for 180 s is *frozen* (working sensors always jitter); faulty signals are blanked before the detectors see them, and the model was trained with the matching `_ok` flags.

8 Behaviour

8.1 Key scenario, dynamic diagram

The demo fire from Section 2, as the containers collaborate at runtime. The diagram exposes what the static views hide: the doors and the evacuation start at the *smoke alarm*, minutes before the fire is confirmed, and the sprinkler command is only "done" when the actuator gateway confirms that BuildSim accepted it.

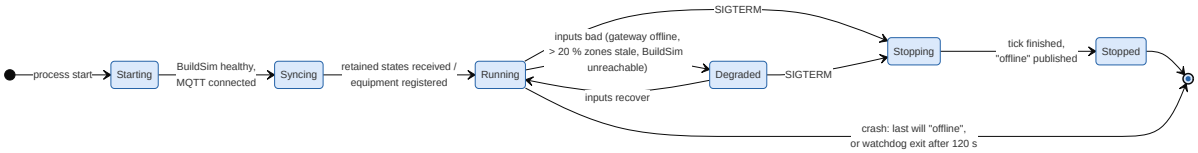


C4 Dynamic diagram, numbered runtime interactions for the demo fire in A109.

8.2 Process lifecycle, state machine

Every service runs the same lifecycle (`lifecycle.py`), published retained on its status topic and as `service_state` events. *Degraded* is where the fault handling lives: the agent keeps every running sprinkler on and raises a critical alert; a sensor gateway that cannot read BuildSim stops publishing (rather than sending stale values with fresh timestamps), so the agent sees those zones offline. A heartbeat file feeds the Docker health check (30 s); a watchdog exits a hung process

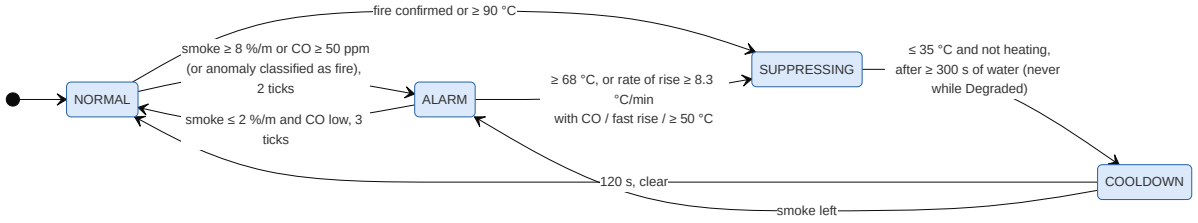
after 120 s so the restart policy restarts it; a crash is announced by the broker through the last will.



Service lifecycle, every container.

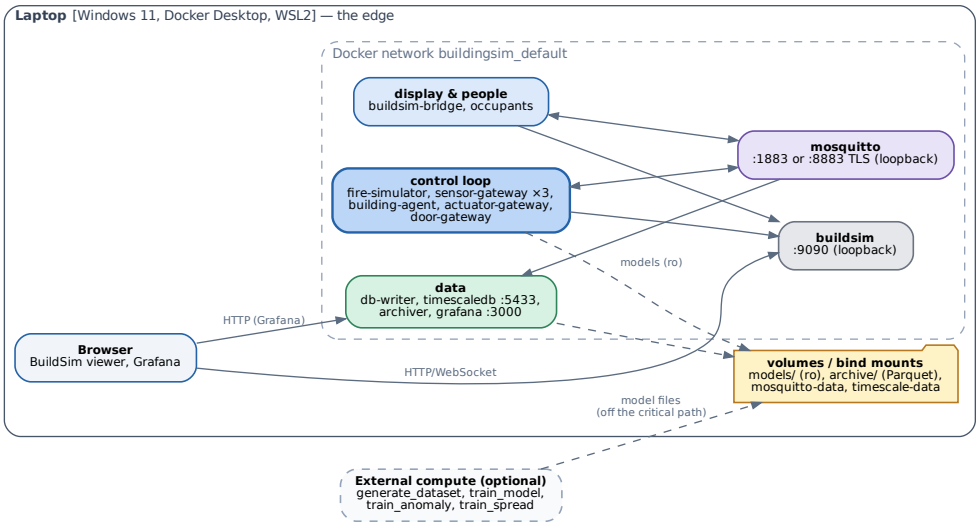
8.3 Zone state machine

Each zone's state in the agent. Every transition is code; the conditions are debounced over two ticks, and suppression can only end through COOLDOWN.



Per-zone state machine of the building agent.

9 Deployment and component placement



Deployment: every container on one laptop (Docker Desktop on WSL2), published on loopback only; training is the only work that could move off the machine.

Everything runs as containers on one laptop, so deployment is about how the system is composed. **Fault isolation**: physics, devices, decisions, actuation, display, people and storage fail indepen-

dently; if the bridge, Grafana or the database dies, detection and suppression continue; if a sensor gateway dies, one floor goes dark and is alarmed by its neighbours; if the agent dies, sprinklers hold their state and the retained plan keeps the evacuation going. **Resources**: the simulator and the agent need CPU, the database disk, the bridge is limited by BuildSim's request rate, not CPU. **Exposure**: only BuildSim (9090), the broker (1883, or 8883 with TLS), the database (5433) and Grafana (3000) are published, all on loopback; secrets come from `.env`; models are mounted read-only. **What merging would lose**: merging the actuator gateway into the agent would remove the independent check on commands; merging the gateways would turn one floor's crash into a building-wide blackout; merging db-writer into the agent would make a slow disk delay a sprinkler. **Edge–cloud**: in a real building the broker, gateways, agent and actuator gateways belong on an edge server in the building, so the loop survives an internet outage; data generation and training (minutes of CPU, off the critical path) could run in the cloud and deliver model files.

10 Test plan

Tests run at three levels: 98 unit tests (physics, sensors, health, features, agent logic, doors, plan, spread, anomaly, clock, schedule, evacuation), 21 end-to-end and fault-injection tests that start a real Mosquitto broker with the production users and ACL, a fake BuildSim that validates like the real one (doors, coverage, highlights), and every service as a real process; and a live run of the full Docker stack on the real BuildSim (Runs A–D, Section 11). `docker compose --profile test run --rm tests` runs the suite in Docker (116 passed, 3 TLS tests skipped there for lack of a broker in the image; they pass locally and TLS was verified in Run D).

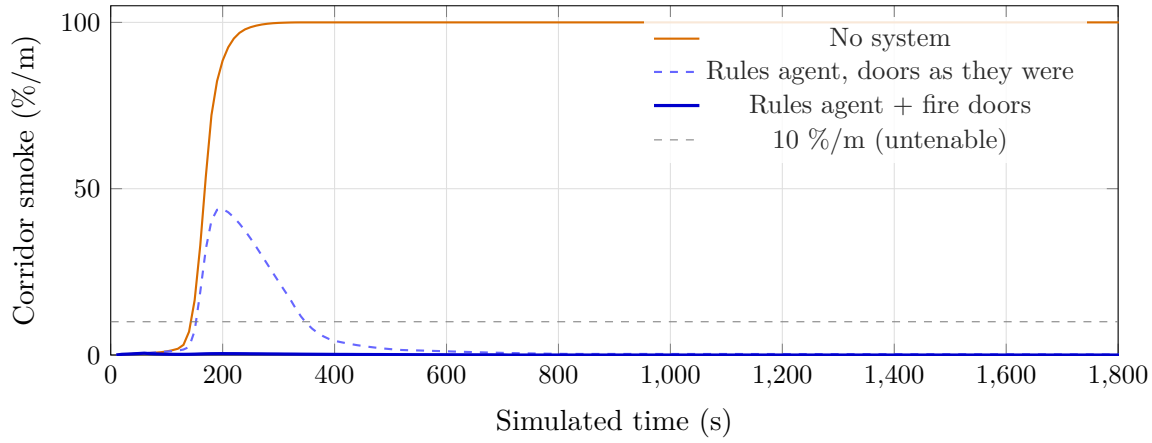
Requirement	Test approach	Evidence
FR-01, FR-03	Closed loop through the fake BuildSim; 150 random scenarios offline; live demo fire	e2e result (release 88 simulated s after flaming); `results/baseline`; live latency
FR-02	Nuisances, neighbouring fires, faults in 150 scenarios; held-out event metrics	False releases per policy (§11)
FR-04, FR-07	Demo fire with and without fire doors	Corridor smoke over time (§11)
FR-06, FR-15	Jam a door during an evacuation; compare routes with BuildSim	`evacuation_replan`, `route_check` events (14/14 agree)
FR-08	Commands that violate the door rules	`door_command_rejected` events
FR-10	Night-time clock, fire alarm	765 doors unlocked (e2e); live Run B screenshots
FR-11, FR-12	Held-out scenarios; hybrid live	`results/anomaly`, `results/spread`; live Run C log
FR-13, FR-14	Plan phasing unit tests; e2e plan, highlights and coverage	Plan versions; no rejected BuildSim update; live screenshots
NFR-01	Agent step for 1×–16× the building	Latency-vs-zones curve (§11)
NFR-02, NFR-04, NFR-08	`kill -9` a gateway; restart BuildSim; two gateways offline	Recovery times (e2e, live)
NFR-03	Restart the agent during suppression; stale retained "on"	Sprinkler adopted, stale state corrected
NFR-05	Malformed, NaN, out-of-range, future, duplicate messages	`data_quality` counts
NFR-06	Live run, database timestamps	`latency.py` (§11)
NFR-07	60 fires with a frozen or offline fire-room thermometer	`results/faults`
NFR-09	SIGTERM; hung loop	"offline" published; watchdog exit

Requirement	Test approach	Evidence
NFR-10	Live run on TimescaleDB	Compression, retention and aggregate jobs listed
SEC-01, SEC-02	No login; wrong user; TLS with an unknown CA; plain MQTT on 8883	Refused connections and publishes
SAF-01	90 °C with the detector saying “no”	Release in the unit test

11 Evaluation and experiments

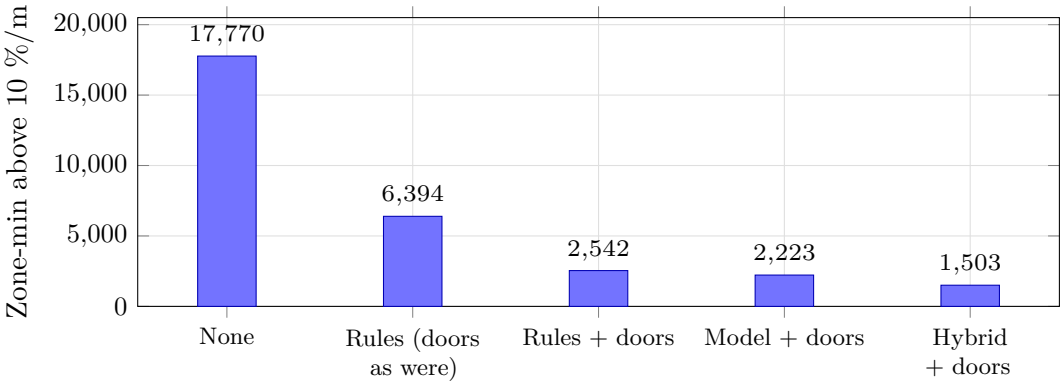
The system was evaluated offline on the same physics, sensors and policy as the live services (`experiments/`), and live on the real BuildSim. The point of this section is evidence, including where the design fails.

One fire over time (FR-04, FR-07). The demo fire in A109, smoke in the corridor outside it. Without the system the room flashes over and the corridor fills completely. The rules agent releases the sprinkler 30 s after flaming; with the doors left as they were the corridor still peaks at 44 %/m and is clear after 890 s; with the fire-door policy the doors close at the smoke alarm and the corridor never exceeds 0.5 %/m, clear after 350 s. Closing doors early does more for the corridor than the sprinkler does.



Many fires (FR-01, FR-02). 150 random scenarios (72 fires, nuisances, quiet periods, faults), every policy on exactly the same scenario.

Policy	Fires released	Release after flaming, median / p90	Flashovers	Smoke, zone-min > 10 %/m	False releases (≈ 970 zone-h)
No system	0 / 72	–	13	17,770	0
Rules, doors as they were	70 / 72	37 / 45 s	0	6,394	2
Rules + fire doors	70 / 72	34 / 44 s	0	2,542	1 (0.1 / 100 zone-h) ✓
LightGBM model + fire doors	72 / 72	−172 / 19 s	0	2,223	4 (0.4) ✓
Hybrid + fire doors	72 / 72	−171 / 31 s	0	1,503	2 (0.2) ✓

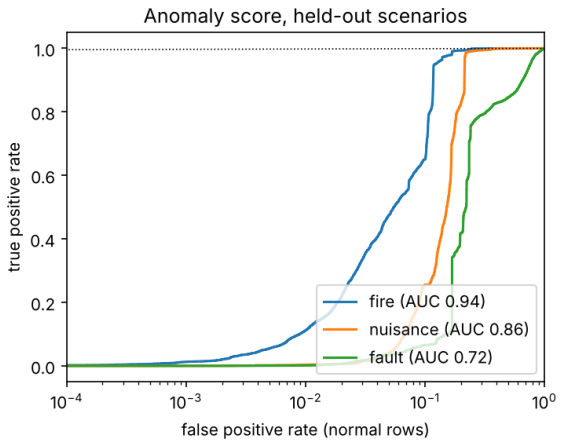


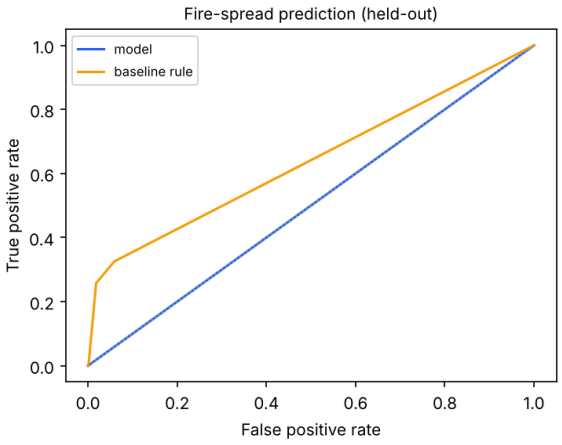
The learned detectors release *before* flaming in most fires (negative delays): they recognise the smouldering signature that the heat rules must wait out, and that cuts the smoke by 40 % (hybrid against rules). The two fires the rules did not release never flashed over.

A broken sensor in the fire room (NFR-07). 60 fires with a frozen or offline thermometer in the burning room before ignition: no system lets 4 flash over; rules, model and hybrid release 60, 59 and 60 of them with one flashover each. The price is precaution: 10 (rules), 31 (model) and 21 (hybrid) releases in zones that never burned, mostly blind rooms and their neighbours. Before the blind-zone rule, closed fire doors hid such fires and 5 of 25 flashed over, the clearest case of one improvement (doors) creating a failure elsewhere.

Fixing a false-release problem (FR-02). The first evaluation with fire doors gave 11 false releases (1.1 per 100 zone-hours, just over the limit). Ten were cooking: once the alarm had closed the doors, the kitchen warmed by 9–12 °C/min at ~5 ppm CO and tripped the rate-of-rise rule. Flaming fires rise faster than 14 °C/min and show $\geq$ 10 ppm CO, so a rate of rise now needs evidence of combustion; false releases fell to 1 with every fire still released in time.

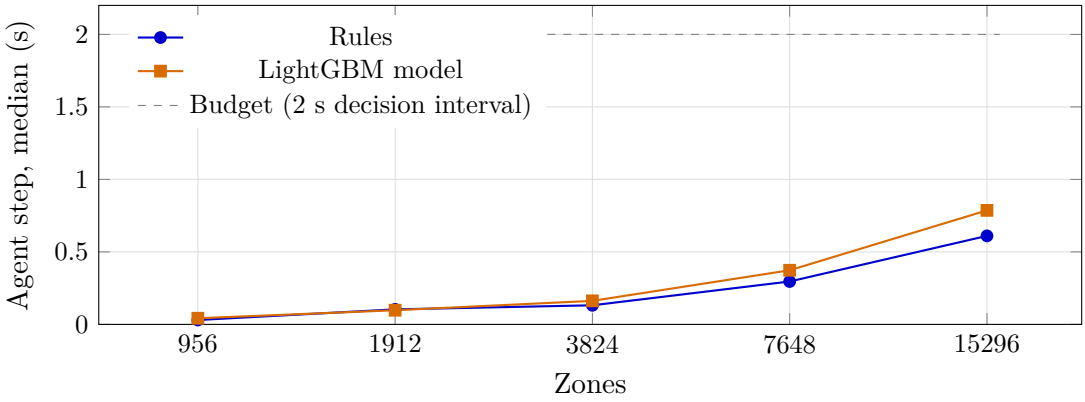
Anomaly detection and spread prediction (FR-11, FR-12). On held-out scenarios every fire and nuisance became anomalous within 60 s (median 10 s), with 0.45 false anomalies per 100 normal zone-hours. Sensor faults are the weak spot: 19 % within 60 s, 88 % within 5 minutes, because drift and a stuck reading develop slowly. Cause accuracy: normal 99.5 %, fire 89.9 %, nuisance 92.9 %, fault 79.7 %. The spread model reaches AUC 0.88 against 0.67 for the neighbour baseline and flags 68 % of the zones that become untenable at least 60 s ahead (baseline 38 %), at precision 0.69 and recall 0.46.





Scaling and the breaking point (NFR-01). The agent's decision time grows linearly with the number of zones: 0.03 s for the building, 0.61 s for 16 copies (15,296 zones), crossing the 2 s interval at roughly 50,000 zones by extrapolation. The real breaking point is elsewhere:

BuildSim takes one HTTP request per sensor value, and mirroring every reading would need 1,912 requests/s for one building, against the bridge's 150/s cap. The deadbands (forward only changes larger than the sensor noise) keep the viewer a few seconds behind during a fire, while the bus and the database have every value.



Live run on the real BuildSim. The whole stack ran in Docker on the real BuildSim on 8 October 2026: Run A (daytime fire with occupants, a sensor gateway killed for 30 s, BuildSim stopped for 30 s), Run B (night, locked doors), Run C (hybrid detector), Run D (TLS).

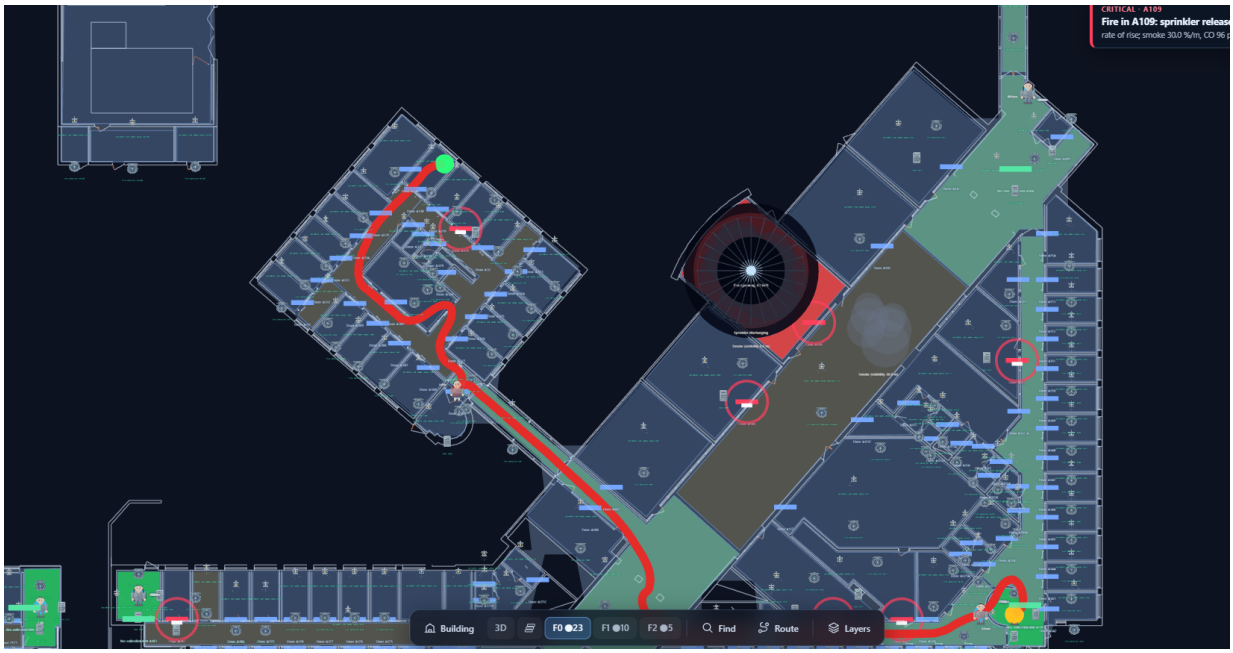
Measurement (live)	Result	Requirement
Flaming → sprinkler command, demo fire	5.8 s wall (29 simulated s)	FR-01 ✓
Sprinkler applied → heat release halved	6.2 s wall	FR-01
Command → accepted by BuildSim and confirmed	sprinkler and damper ≤ 0.01 s; floor doors 1.0 s median (p90 1.1 s, 227 commands); lifts 1.0 s	NFR-06 ✓
Sensor gateway killed 30 s → zones back	median 14 s, p90 31 s after the kill (includes the outage)	NFR-02 ✓
BuildSim stopped 30 s	services Degraded, then Running; equipment, doors and states restored	NFR-04, NFR-08 ✓
Storage	compression, two retention and the aggregate jobs scheduled on TimescaleDB	NFR-10 ✓
Night alarm (22:00)	locked doors released at the alarm, back to the schedule at the all-clear	FR-10 ✓
Hybrid	alarm from the anomaly stage (P = 0.74), water from the cause classifier (P = 1.00) before the heat limits	FR-11 ✓
TLS	every service connected on 8883, all healthy	SEC-02 ✓

What the live run found. The offline tests had not caught six defects, all fixed and now covered by tests. (1) BuildSim's viewer flies the camera to every route it receives, and a new route went out every 5 s, so the operator could not look at the floor: one route per evacuation is now sent by default. (2) Spread risk was reported all over the building and changed every few seconds; each change re-planned every route and reset the stair timer, so 7 of 8 people never got downstairs: risk is now limited to three links from an alarm, spread-only re-plans to one per 10 s, and stair progress survives a re-plan. (3) A few upper-floor rooms extend past the floor page and BuildSim rejected the whole people update: positions are kept on the page. (4) A retained "sprinkler on" from a previous run survived `docker compose down` and was adopted at start-up: the actuator gateway now takes the truth from BuildSim at start and corrects stale confirmations.

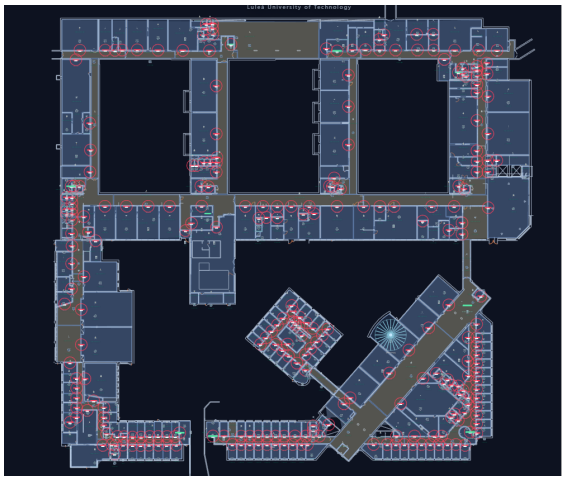
(5) The day/night and detector settings were fixed in the compose file. (6) The route cross-check found nothing to compare at night, when people start in locked offices. None of these was visible against a fake BuildSim; all were obvious within minutes on the real one. That is the strongest argument for the live run.

12 Dashboard

The **BuildSim viewer** is the operational dashboard. From the agent's plan the occupants service highlights burning zones **red**, smoke and predicted spread **orange** (lighter for risk), and the exits and the corridors on the escape routes **green**; rooms with people awaiting rescue are purple. The bridge draws smoke spread as **coverage zones** from the *measured* smoke (grey to near-black as it thickens), so the operator sees what the system sees, not the simulator's truth. One evacuee's remaining route is shown; the alert list carries the agent's reasons ("Fire in A109: sprinkler released, rate of rise; smoke 30 %/m, CO 96 ppm").



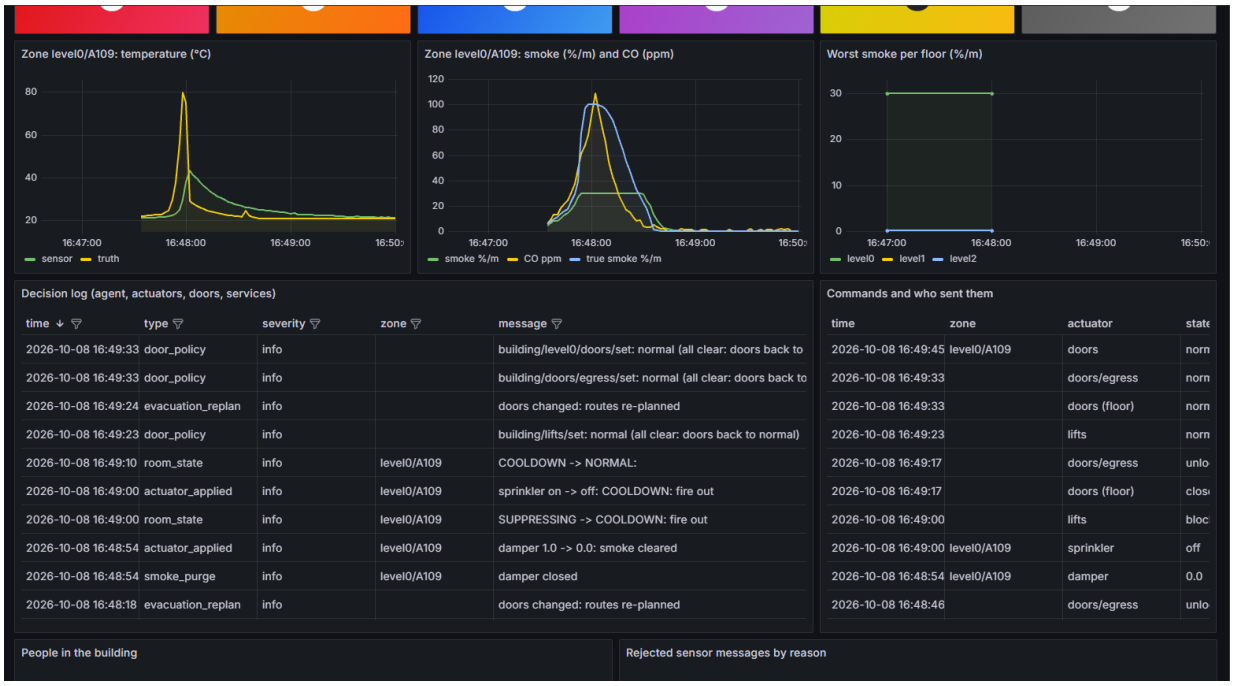
Live run A on the real BuildSim: A109 red with its sprinkler and smoke, escape corridors and exits green, one evacuation route, people on all floors (F0: 23, F1: 10, F2: 5).





Grafana gives what BuildSim cannot, because BuildSim keeps no history: status tiles (fires, alarms, sprinklers, dampers, degraded services, rejected messages), sensor against truth for a selected zone, the worst smoke per floor, people in the building over the day, and the **decision**

log, every state change with its reason, every command with who sent it. The decision log is what makes an autonomous system trustworthy enough to leave running: an operator sees not just what it did but why.



Grafana during live run A: temperature, smoke and CO in A109 (sensor against truth), the decision log and the commands with their source.

13 Risks and critical reflection

The system is only as good as its simulator. Every model, threshold and result here was obtained against physics we wrote; the rules' thresholds are real-world values, but the learned models have learned *our* correlations (for example, that flaming fires produce CO early), and a real building would need real data before they could be trusted. That is the main reason the rules stay the default and every model sits behind the safety net.

The fire doors were the most effective single measure (smoke exposure down by 60 %), and also the source of two new failure modes: a fire hidden in a room with broken sensors, and cooking that looked like a fire once the kitchen door closed. Both were found only because the evaluation ran the full closed loop over many scenarios. The fixes (blind-zone precaution, combustion evidence) cost precautionary water in rooms that never burn when sensors are broken: we chose water damage over flashover, and that trade-off is deliberate.

The learned components have honest weaknesses: sensor faults become anomalous only slowly; spread prediction misses about half of the onsets at its operating point and was over-confident far from any fire until limited to three links. The evacuation model assumes people follow instructions after a pre-movement delay; real crowds queue at stairs and doors, which our routing ignores. Finally, the live run showed that the integration with the real BuildSim is where the surprises are: six defects in one afternoon, none of which the fake BuildSim could reveal.

Risk / assumption	Impact	Mitigation / decision
Models learned the simulator, not reality	Wrong decisions on real data	Rules default; safety net; retrain on recorded live data (`export_dataset.py`)

Risk / assumption	Impact	Mitigation / decision
Slowly drifting sensor	Late anomaly, late fallback	Health checks + anomaly stage; drift detection over hours would be next
Broker is a single point of failure	Readings and commands stop	Retained state and last wills; actuators hold state; restart policy; a second broker would be next
BuildSim's one-request-per-value API	Viewer lags in a large fire	Deadbands and rate cap; the bus and database have every value
Fire doors close for nuisance alarms	Inconvenience; smoke still contained	Doors return 120 s after the last alarm
Crowd behaviour not modelled	Evacuation times optimistic	Report times as lower bounds; add queueing at stairs
Precautionary releases with broken sensors	Water damage in rooms that never burn	Deliberate: water over flashover (NFR-07)
Development passwords in the repository	Unauthorised commands if exposed	`.env` secrets, TLS listener; broker on loopback only

What we would do next: record a long live run and retrain the models on it; detect slow drift; model queueing in the evacuation; and replace the display mirror's per-value requests with a batched BuildSim endpoint if one became available.